

Solution Requirements

Date: 29 September 2026 **Team ID:** SWTID-2026-4886
Project: AnswerBot – AI-powered FAQ and Customer Support Automation

Functional Requirements

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Register and log in users using email/password; passwords are hashed with bcrypt.	High
2	Authorization	Protect FAQ modification and AI routes using JWT bearer authentication.	High
3	External Interfaces	Integrate Google Gemini for AI answer and FAQ generation; connect MongoDB Atlas through Mongoose.	High
4	Transaction Processing	Validate request payloads, execute CRUD/search operations, call AI service when requested, and return JSON status responses.	High
5	Reporting	Expose FAQ retrieval/search results and structured API responses for client applications.	Medium
6	Business Rules	Only authenticated users may access protected operations; FAQ records retain creator reference; invalid requests return controlled errors.	High
7	Compliance / Security	Protect credentials with hashing; keep secrets in environment configuration; do not expose API keys in source code.	High

Non-Functional Requirements

NFR	Requirement
Security	JWT route protection, bcrypt password hashing, environment-based secrets.
Reliability	Centralized error handling and validation for critical operations.
Performance	API should respond efficiently under normal development/demo load; formal load benchmarks are not claimed unless measured.
Maintainability	Modular routes, controllers/services, models and middleware.
Scalability	MongoDB-backed stateless API can be deployed behind a scalable server architecture.
Usability	Clear JSON response structure and API testing through Postman/Thunder Client.